AA-R-001691(GSA-FSS)
December 29, 1972

INTERIM FEDERAL SPECIFICATION

RACK, STORAGE, SOLID SHELF, GENERAL PURPOSE

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Washington, D. C., 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers solid shelf storage racks for storing non-palletized, packaged items.

1.2 Classification. The storage racks shall be of one type, and of the following styles, classes, depths, and heights:

Style A. - Starter unit. 72-inch width shelves with rated capacity of 1500 pounds.
Style B. - Add-on unit. 72-inch width shelves with rated capacity of 1500 pounds.

Class 1. - 10,000 pounds (minimum) rated capacity upright frame
Class 2. - 16,000 pounds (minimum) rated capacity upright frame
Class 3. - 20,000 pounds (minimum) rated capacity upright frame

Depth A. - 18 inches
Depth B. - 24 inches
Depth C. - 30 inches
Depth D. - 36 inches
Depth E. - 48 inches

Height 1. - 72 inches
Height 2. - 96 inches
Height 3. - 120 inches
Height 4. - 144 inches
Height 5. - 168 inches
Height 6. - 192 inches
Height 7. - 216 inches
Height 8. - 240 inches

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

TT-C-490 - Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings.
PPP-C-650 - Crates, Wood, Open and Covered.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).
Fed. Std. No. 595 - Colors.

FSC 7125

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, MO., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-P-116 - Preservation, Methods Of.
MIL-W-12332 - Welding, Resistance, Spot, Seam, and Projection; for Fabricating Assemblies of Low-Carbon Steel.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U. S. Military Property.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Welding Society (AWS) Publication:

A3.0-69 - Terms and Definitions.

(Application for copies may be addressed to American Welding Society, 33 West 39th Street, New York, New York 10018.)

American Institute of Steel Construction Publication:

Specification for the Design, Fabrication and Erection of Structural Steel for Buildings.

(Application for copies may be addressed to the American Institute of Steel Construction, 101 Park Avenue, New York, New York 10017.)

American Iron and Steel Institute Publication:

Specification for the Design of Cold-Formed Steel Structural Members.

(Application for copies may be addressed to the American Iron and Steel Institute, 150 East Forty-Second Street, New York, New York 10017.)

National Motor Freight Traffic Association, Incorporated, Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, D. C. 20036.)

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Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606).

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.2), before production is commenced, a sample of the finished commodity and each component part shall be submitted or made available for the contracting officer or his authorized representatives to examine and test to determine compliance with this specification. Approval of the preproduction sample authorizes the commencement of production, but does not relieve the contractor of the responsibility of complying with all other applicable provisions of this specification. Production units shall not vary from the approved preproduction sample in design and construction without the written approval of the contracting officer.

3.2 Materials. All materials used in the construction of the racks shall be as hereinafter specified. Any material permitted by the specification, that is not covered by the subsequent material paragraphs, shall be a commercial material. All material shall be unused (not previously used in the present state) and free of all imperfections that affect appearance or serviceability of the finished commodity. All materials shall meet the applicable test requirements specified in section 4 of this specification.

3.2.1 Steel. The steel used for the fabrication of the storage racks shall be in accordance with one of the American Society for Testing and Material (ASTM) designations for structural steel referenced in the Specification for the Design, Fabrication and Erection of Structural Steel for Buildings, or one of the suitable ASTM designations for structural quality sheet or strip referenced in the Specification for the Design of Cold-Formed Steel Structural Members. The steel shall be of a designated grade, referenced in the respective ASTM specification. The contractor shall provide certification that the steel used for the racks meets the respective ASTM specification. Alternatively, the steel may be a high strength, low-alloy steel that meets or exceeds the mechanical requirements of one or more of the steels delineated in the ASTM specifications of AISI or AISC. The contractor shall provide the chemical composition, yield point data, and per cent of elongation of the test sample of the alternate steel. The steel used shall be of sufficient mass to assure compliance with all test requirements of this specification.

3.2.2 Finishing material.

3.2.2.1 Primer. The primer (undercoat) when used (see 3.7.2), shall be a commercial product that is compatible with the selected finishing enamel.

3.2.2.2 Enamel. The finishing enamel shall be air-drying or a baking type, selected from an enamel manufacturer's commercial products. The enamel finish shall be in accordance with 3.7.3, and the color of the finish shall be in accordance with 3.2.2.3.

3.2.2.3 Color of finish. Unless otherwise specified (see 6.2), the color of the finish shall be green, color number 24277 of Federal Standard No. 595. Unless otherwise specified, the color of the safety lock shall be the color of the rack, or shall be commercial finished cadmium, chromium, or zinc plate. The color of the finish shall be closer to the specified Fed. Std. No. 595 color than to any other Fed. Std. No. 595 color with the same first two digits. (For example, color no. 24277 shall be closer to color chip 24277 than any other Fed. Std. No. 595 color with the same first two digits of 24). When necessary, test method 4250 of Fed. Test Std. No. 141 shall be utilized to make the visual color comparisons. Sample color chips of the standard enamel colors are obtainable from the Business Service Center, Federal Supply Service, General Services Administration, Washington, D. C., 20407, or from the Business Service Center of the nearest GSA Regional Office.

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3.3 Design. The basic design of the package storage racks shall be similar to figure 1. The design of all steel load bearing components, hot-rolled to shape, shall be in accordance with the Specification for the Design, Fabrication, and Erection of Structural Steel for Buildings, except as specified herein. The design of all steel load bearing components, cold formed to shape shall be in accordance with the Specification for the Design of Cold-Formed Steel Structural Members, except as specified herein. The design and construction of each load bearing component shall assure that the component is capable of sustaining the specified minimum safe load, and the applied load required to establish the specified minimum safety factors.

3.3.1 Safety factor. The safety factor, for the purpose of this specification, shall be established by the ratio of the test loads (see section 4) to the specified minimum load for the respective components. The test load for the shelf components shall be 1.65 of the specified, minimum capacity (see 1.2). The test load for the upright frame shall be 1.95 of the specified, minimum capacity for each class upright frame (see 1.2).

3.4 Welding. The parts joined by welding shall be by the specific welding process specified hereinafter, or if not specified, by one or more of the welding processes specified in the sub-paragraphs below. All welds shall be properly fused, sound, neatly accomplished, free of cracks, pits, flash and spatter. All rough or sharp welds shall be finished smooth to prevent personal injury.

3.4.1 Arc welding. Arc welding shall be accomplished by a suitable arc welding process referenced in AWS A3.0-69. Arc welds shall not be applied over spot or projection welds.

3.4.2 Projection, resistance, seam, and spot welding. The projection, resistance, seam, and spot welding shall be in accordance with MIL-W-12332. The spot welds shall be located not closer to the edge of either adjoining member than one-half the greatest dimension of the spot weld. The spot and projection welds shall be spaced on centers that will provide adequate strength for the welds and welded component(s) to meet the tests specified in section 4. The welding process shall be controlled to produce uniform welds that will meet the visual examination and peel test specified in sub-paragraphs of 7.3 in MIL-W-12332. All projection and seam welds shall meet the visual examination and the test specified in the applicable sub-paragraphs of 7.3 of MIL-W-12332.

3.5 Dimensions. The dimensions of the racks and components shall be as specified herein, and depicted in figure 1. Any dimension not specified shall be as established by the manufacturer.

3.5.1 Tolerances. The dimensions specified herein with a referenced tolerance shall be as specified. All dimensions specified herein without a referenced tolerance shall have the following tolerances:

Any component dimension (other than overall dimensions), machined to size - plus or minus 1/64 inch; roller to size - plus or minus 1/32 inch.
Overall component dimension - plus or minus 1/16 inch.
Overall unit depth (see 1.2) - plus or minus 1/8 inch.
Overall unit height (see 1.2) - plus or minus 3/16 inch.

The referenced tolerances may be cumulative but not to the extent that critical, specified, or overall tolerances cannot be maintained, or the components and sub-assemblies will not easily assemble.

3.6 Construction. The construction shall be as specified in the following sub-paragraphs. The construction of all components not specified herein shall be in accordance with good engineering practices. The components shall be constructed to produce a rigid, stable rack that will withstand all tests specified in section 4. The racks shall be capable of being assembled side by side in a row, and connected back to back in a double row.

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3.6.1 Upright frame. The upright frame shall be constructed of the steel specified in 3.2.1, and shall be of the class, depth, and height specified in 1.2 (also see 6.2). The frame shall be all welded construction of post, diagonal and horizontal braces (struts) and floor bearing plate. The upright frame shall be a continuous, one piece assembly for heights 1 (72 inches) thru 5 (168 inches), and for heights 6 (192 inches) thru 8 (240 inches), the upright may be in one or two sections. All connecting hardware shall be provided with the two-section upright. The connection (splice) shall not prevent the location of the shelf beam or tie beam at the point of post intersection. The width of the post (facing the aisle) shall be not less than 1-5/8 inches or more than 3 inches. The bottom of each post shall have a floor bearing plate securely welded thereto. The plate shall be of not less than 11 gage (0.1196 inch) steel. The plates used on all class 1 upright frames shall have not less than 7.5 square inches of bearing surface, and the plates used on all class 2 and 3 upright frames shall have not less than 12.5 square inches of bearing surface. The attached plate shall completely cover the end of the post, and have an accessible hole, not less than 1/2 inch diameter, for anchoring the upright frame to the floor. Each post shall have a single or double row of openings commencing not farther than 6 inches from the bottom of the floor bearing plate and continuing upward to the top of the post by either 2, 3, or 4 inch increments. The openings shall readily accommodate two shelf beam connectors or two tie beam connectors, end to end, at corresponding heights throughout the height of the post. Each post, of the units with separate shelf supports, shall have an additional row (or rows, as applicable) of openings for attaching the support. The increments shall be the same as used for the shelf beam or tie beam. All openings of one post shall be identical to the openings of all other posts of the same class of upright frames. The location of all corresponding openings of the posts of any one style, class, depth and height upright shall not vary more than 3/64 inch from the top of the floor bearing plates to 5 feet on the posts, 3/32 inch to 10 feet on the posts, 9/64 inch to 15 feet on the posts, and 3/16 inch to 20 feet on the posts. The arrangement of the openings shall not prevent the attachment of shelves on the same elevation within each bay. The posts shall be braced by horizontal and diagonal or "X" braces as illustrated in figure 1. The upright frame shall have sufficient number of diagonal and horizontal braces to produce a stable upright that will pass the tests specified in section 4. The braces shall not protrude beyond the sides of the post, obstruct the attachment of any component to the post, or interfere with the loading of the shelves.

3.6.2 Shelves. The shelves shall be constructed of not less than 16 gage (0.0598 inch) steel specified in 3.2.1. The clear shelf width (distance between upright posts) shall be 72 inches, plus 1/4 inch, minus 1/8 inch. The shelf depth shall be plus or minus 1/8 inch of the depth of the upright frame (see 1.2). The exposed beam width shall be considered as shelf depth for flush shelves that are supported on beam ledges, or in beam openings. Each shelf shall have a minimum safe load capacity of 1500 pounds. The shelves shall be supported by shelf beams that connect to, and stabilize the uprights; or by end supports that connect to the uprights (see figure 1). The shelves shall be in not more than three sections and the shelf sections shall be constructed to form a continuous, full length and width shelf with not more than 3/64 inch gap between the sections. The assembled shelves (including sections) shall be flat to within 3/8 inch. The shelves, supported by shelf beams, shall lap the top of the beam and extend downward over the sides of the beams (front and rear) not less than 1/2 inch, with not more than 1/16 inch gap (front and rear) between the shelf and the beams. Alternatively, the shelves may locate on beam ledges or position in beam openings. The top of the shelves of this alternate construction shall be flush, (plus or minus 3/64 inch), with the top of the beams. The shelf construction shall be the same for all units on any one order or contract. All shelves shall be restricted from vertical movement in excess of 7/16 inch, and all shelves shall be restricted from lateral movement in excess of 5/16 inch. The shelves shall be constructed to withstand the tests in section 4.

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3.6.3 Shelf and tie beams. All units shall have either shelf beams or tie beams to stabilize the uprights. The beams shall establish the bay width of the unit. The bay width shall be not less than the specified minimum shelf width. The beams shall have boltless, multiple prong or stud type connectors, at each end, that will slip-fit into the openings in the upright posts. Each end of the beams shall be equipped with a beam-to-post safety lock specified in 3.6.3.1. Two (one pair) shelf beams shall be furnished with each shelf, and four (two pair) tie beams shall be furnished with heights 1 (72 inches) thru 4 (144 inches) units. Six (three pair) tie beams shall be furnished with heights 5 (168 inches) thru 8 (240 inches) units. One pair of tie beams and shelf beams (supporting an unloaded shelf), of an assembled and otherwise loaded rack, consisting of three or more pairs of tie or shelf beams, shall be capable of being removed and repositioned without dismantling the rack or employing the use of special tools when tested in accordance with 4.5.5. The beams and beam-to-post connections shall comply with the test specified in section 4.

3.6.3.1 Beam-to-post safety lock. Each beam-to-post connection shall have an automatic or manually operated safety lock that is effective throughout the range of the beam connections. The safety lock shall prevent disengagement of the beam-to-post connection when the uprights, beams, or shelves are subjected to lateral or vertical impact. The lock may be an integral part of the beam or may consist of additional components. The safety lock shall engage and disengage without the use of special tools. The locking mechanism shall not obstruct or restrict the loading or unloading of the shelves, or the repositioning of the beams. The safety lock shall withstand the test specified in 4.5.6.

3.6.4 Shelf end supports. The end supports shall be constructed to receive and adequately support the specified minimum shelf load and the test load specified in 4.5.4. Each end of the support shall attach to the upright post by two or more boltless fasteners. Each post attachment shall have an automatic or manually operated locking device to prevent disengagement of the support when tested in accordance with 4.5.6. The supports shall restrict the lateral movement of the shelf in excess of 5/16 and vertical movement of the shelf in excess of 7/16 inch. The supports shall not restrict or interfere with the loading or unloading of the shelves. The supports, supporting an unloaded shelf of an assembled and an otherwise loaded rack, shall be capable of being removed and repositioned without dismantling the rack or employing the use of special tools. The end supports shall comply to the test specified in section 4.

3.6.5 Back-to-back connector. When specified (see 6.2), back-to-back connectors shall be furnished with the rack units. The connectors shall be either the fixed or adjustable type and shall connect to the upright posts to form a double row arrangement. Unless otherwise specified (see 6.2), the connectors shall provide 4-inch spacing between the connected posts. Each end of the connectors shall securely attach to any post of the upright frames, at not less than two locations on the lower and the upper half of the post by bolts, washers, and nuts. Alternatively, the connector attachment may be boltless, at no less than two locations within the lower and upper limits of shelf attachment. The boltless attachment shall have a safety locking device to prevent accidental disengagement. The connectors shall be of sufficient size and strength to withstand the tests specified in section 4.

3.6.6 Wall-to-post connector. When specified (see 6.2), wall-to-post connectors shall be furnished with the rack units. The connector shall be adjustable to compensate for wall variations, or alternatively, the connector may be the fixed type and supplied with shims to compensate for wall variations. Unless otherwise specified (see 6.2), the connector shall provide 4-inch spacing between the post of the upright and the wall. The connector shall have not less than two holes of not less than 7/16 inch diameter, for attaching the connector to the wall. The connector shall withstand the tests specified in section 4.

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3.6.7 Leveling shims. When specified (see 6.2), leveling shims shall be provided with the rack units. The thickness of the shims shall range from 10 gage (0.1345 inch) thru 16 gage (0.0598 inch). The shims shall be of the same configuration and size as the floor bearing plate. The shims shall have a hole corresponding to and aligning with the hole in the floor bearing plate. The shims shall have a coating of all lacquer, enamel, or a metallic coating to resist corrosion.

3.7 Pretreatment and finishing.

3.7.1 Pretreatment. All surfaces to receive the enamel finish shall be smooth, free of burrs, flash, spatter, sharp edges, and rough or sharp welds before pretreatment. All surfaces, exclusive of plated safety lock, shall be prepared for painting in accordance with any cleaning method of TT-C-490.

3.7.2 Priming. All precleaned surfaces shall have a coating of primer specified in 3.2.2.1, unless the surfaces are prepared for finishing with a chemical conversion coating, in accordance with type I, type II, or type IV of TT-C-490. The primer shall be applied and dried as soon as possible after the cleaning operation.

3.7.3 Enameling. All surfaces prepared for finishing as specified in 3.7.1 and 3.7.2 and shall be coated with the enamel specified in 3.2.2.2. The total dry film thickness of the primer and enamel shall be not less than 0.9 mils on all exposed (exterior) surfaces and the dry film thickness of the enamel on exposed (exterior) surfaces prepared for finishing in accordance with type I, type II, or type IV, (chemical conversion and pretreatment coatings) of TT-C-490 shall be not less than 0.6 mils. The color of the finish shall correspond to the color number referenced in 3.2.2.3. The finish shall level out to produce a well bonded, smooth, hard surface, and shall be free of runs, foreign matter, areas of thin or no film, and separation of color. The enamel finish shall conform to the tests specified in 4.5.9.

3.8 Assembly. The components of the rack shall assemble and disassemble, easily and freely, without the use of special tools. The tie beams, shelf beams, and end supports shall assemble and disassemble with the upright frames, throughout the height range of the post openings. Adjacent shelves shall assemble with the common upright, at corresponding heights throughout the height range of the post openings. The safety lock shall engage the locking component and disengage without the use of special tools. An assembled rack shall be rigid and stable to withstand the tests specified in section 4.

3.9 Identification marking. Unless otherwise specified, each upright frame, tie beam and shelf beam shall be permanently and legibly marked, using permanent ink and stencil or rubber stamps, pressure sensitive label or decal, engraved, or die stamped with the contract number, and the manufacturer's name or trade-mark, so that the source of supply may be readily determined. The rated capacity of the upright frames shall be conspicuously marked on each upright post, not less than 4 feet or more than 6 feet from the floor bearing plate. For example, class 1 uprights shall be marked "Rated capacity - 10,000 lbs.", etc. When specified (see 6.2), the components shall be marked for identification in accordance with MIL-STD-130. The rack components may be marked with the manufacturer's commercial marking, in addition to the above required marking.

3.10 Workmanship. The workmanship shall produce a serviceable product acceptable under the requirements specified herein. All welds, edges, and corners shall be free of sharp projections and burrs. Formed parts shall be uniform, without cracks, fractures, and undesirable tears and holes. All openings and holes shall be uniform and within specified tolerances. All referenced dimensions shall be within specified tolerances. All movable parts shall operate and function properly. Attention shall be given to pretreatment operations to assure that all surfaces are pretreated and the proper sequence of operations is maintained and completed. The finish shall be applied in accordance with good commercial practice to assure complete coverage, specified dry film thickness, and durability of the finish. The workmanship shall assure that components will assemble and disassemble as specified in 3.8, and the occurrence of defects will not exceed the acceptable quality level specified herein.

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3.11 Assembly instructions and parts pamphlet. An illustrated assembly instruction sheet and a manufacturer's parts pamphlet shall be supplied with each order of racks or components. The assembly instructions shall be printed on durable paper, in English, using letters large enough to distinguish by normal eyesight. The parts pamphlet shall contain an illustration of all components of the rack and reference the manufacturer's commercial or catalog number of each component.

3.12 Conflict of text and figure. In any case of conflict between the text and figure 1, the text shall prevail.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection responsibility. The supplier is responsible for the performance of all inspection and tests requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. In accordance with 4.1 above, the supplier is responsible for insuring that components and materials used are manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or if none, in accordance with this specification.

4.2 Sampling for inspection and acceptance. Except where otherwise indicated, sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105.

4.3 End item inspection.

4.3.1 Visual examination. Examination of the storage racks shall be in accordance with the classification of defects specified in table I and the inspection and acceptable quality levels specified in table III. The lot size shall be expressed in units of racks for the purpose of determining the sample size in accordance with MIL-STD-105. The sample unit for this examination shall be one complete storage rack of each style, class, depth, and height on contract.

TABLE I. Classification of defects

Examine	Defect	Classification Major	Minor
Steel	Steel contains scale, rust, pits, fins, sharp edges, protrusions, burrs or undesirable holes.	X	
	Previously formed and used.		X
	Not in accordance with paragraph 3.2.1.		X
Welds	Welds not neatly accomplished, burned thru, cracked, or poorly fused.	X	
	Rough, sharp projections, flash and spatter not removed.	X	
	Welding method employed other than specified.		X

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TABLE I. Classification of defects (continued)

Examine	Defect	Classification Major	Minor
Upright frames	Frame design not similar to figure 1, or the construction is not in accordance with the class, depth, and height specified in 1.2.	X	
	Openings in post will not accommodate two tie or shelf beam connectors end-to-end.	X	
	Openings in post missing, not spaced in increments specified, or the openings commence farther than 6 inches from the bottom.	X	
	Floor bearing plate is not attached, or size is less than minimum specified.		X
	Hole missing, wrong size, or not accessible in floor bearing plate.		X
Shims	Shims are not of the gage steel specified.		X
	The size and configuration is not identical to the floor bearing plate.		X
	Hole missing, or hole will not align with hole in floor bearing plate.	X	
Back-to back connector	Incapable of providing the specified spacing between the posts.		X
	Will not securely attach to post, or attach to post at two locations within the lower and upper limits of the shelf attachment.	X	
	Boltless attachment not equipped with safety locking device.	X	
Shelves	Shelf sections exceed the maximum number specified.		X
	Shelves are not flat to within 3/8 inch.		X
	Shelves are not restricted from lateral movement in excess of 5/16 inch or vertical movement in excess of 7/16 inch.		X
Shelf and tie beams	The beams do not have boltless, multiple prong or stud end fasteners.	X	
	The beams do not establish the proper bay width.		X
	Beam connectors will not properly position in post openings.	X	
	Shelf beams do not restrict lateral movement of the shelves, in excess of 5/16 inch.	X	

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TABLE I. Classification of defects (continued)

Examine	Defect	Classification Major	Classification Minor
Safety lock	Missing at the specified connections.	X	
	Ineffective or inoporative.	X	
	Restricts the loading or unloading of the shelves.		X
	Special tools required to operate.		X
Shelf end supports	Less than two boltless fasteners on each end of support.	X	
	Support does not restrict lateral movement of shelf in excess of 5/16 inch or vertical movement in excess of 7/16 inch.	X	
	Supports interfere with loading or unloading of shelves.	X	
Pretreatment and finish	Pretreatment omitted, primer omitted when required, and area(s) of no film.	X	
	Poor adhesion, blistering, or peeling.		X
	Finish contains runs, sag, area(s) of thin film, or separation of color.		X
	Color of finish other than specified.		X
	Dry film thickness less than specified.		X
Assembly	Components will not assemble and disassemble without the use of special tools.	X	
	Shelf and tie beams, and end supports will not assemble with posts throughout the range of openings.	X	
	Assembled rack is not rigid and stable when loaded with capacity or test loads.	X	
Workmanship	Missing component, except bolt, washer or nut.	X	
	Missing bolt, washer, or nut.		X
	Malformed or damaged part.	X	
	Burrs, sharp edges or sharp or rough welds.	X	
	Undesirable holes or tears in parts.		X
	Holes and openings not uniform.	X	
	Assembly instructions and parts phamplet are not furnished with each order of racks or components, or the information is not legible.	X	

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4.3.2 Dimensional examination. Inspection will be made of the storage racks for compliance with the dimensions specified herein, including the thickness of metal. Any dimension not within the specified tolerance will be classified as a defect. The inspection and acceptable quality level shall be as shown in table III.

4.3.3 Inspection of preparation for delivery. An inspection shall be made to determine that the packaging, packing and marking comply with the requirements in Section 5 of this specification. Defects shall be scored in accordance with table II. For examination of interior packaging the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipments fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level and acceptable quality level shall be as shown in table III.

TABLE II. Classification of preparation for delivery defects

Examine	Defects
Markings	Omitted; incorrect; improper size; location, sequence, or method of application.
Material	Any component missing or damaged.
Workmanship	Inadequate application of components, such as loose strapping.

4.4 Inspection levels and acceptable quality levels (AQL's). The inspection levels and the acceptable quality levels expressed in defects per hundred units shall be in accordance with MIL-STD-105, and as shown in table III.

TABLE III. Inspection and Acceptable Quality Levels

	Inspection level	Acceptable Quality Level Major	Total
For defects listed in 4.3.1	S-2	4.0	10.0
For defects listed in 4.3.2	S-2	---	4.0
For defects listed in 4.3.3	S-2	---	4.0

4.5 Tests.

4.5.1 Test criteria. The following tests shall be performed on all test units using inspection level S-1 for the selection of the number of racks to be tested from any lot. Failure of any unit to pass all the tests shall be sufficient cause to reject the lot from which the test sample was selected. The lot size shall be expressed in units of racks. Unless otherwise specified, all classes, depths, and heights of racks on contract shall be subjected to the tests. All tests shall be conducted in the presence of the contracting officer or his authorized representative(s). The Government reserves the right to use additional equipment, information or methods

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deemed necessary to determine compliance with the requirements of this specification. Unless otherwise specified, each unit tested shall be a completely assembled two-bay rack consisting of 3 upright frames and 6 (3 per bay) shelves for upright heights 1 (72 inches) thru 4 (144 inches), 8 (4 per bay) shelves for heights 5 (168 inches) thru 6 (192 inches), and 10 (5 per bay) shelves for heights 7 (216 inches) thru 8 (240 inches). The test units that utilize tie beams shall consist of 8 (2 pair per bay) of beams for heights 1 (72 inches) thru 4 (144 inches) and 12 (3 pair per bay) of beams for heights 5 (168 inches) thru 8 (240 inches). The shelves and tie beams shall be attached to the upright frames so that they are uniformly spaced, the lowest beam and shelf is not less than 32 inches from the floor plate, and the top beam and shelf is located where they will be free of any load applied to the top of the upright frame. All loads referenced in the individual test for the upright and shelves shall have a tolerance of plus 75 pounds, minus 25 pounds. However, the minimum allowable load(s) shall not be intentionally used for the tests. The sequence of the tests may be as herein delineated or any sequence of tests favorable to existing circumstances may be used. The maximum calibration of all measuring devices, other than dial indicators, used in conjunction with the tests shall be 0.010 inch. Dial indicators shall be calibrated in increments of 0.001 inch or less.

4.5.2 Spot, projection, and seam weld test. Welded specimens representing spot, projection, or seam welding processes utilized in the construction of the storage racks shall be offered for test. The specimens shall be subjected to the respective test in the sub-paragraph of 7.3 of MIL-W-12332.

4.5.3 Upright load test. The test unit shall be as specified in 4.5.1. The test unit shall not be anchored to the floor or to a wall. The specified load shall be mechanically applied to the top of the posts of the intermediate (common) upright frame. The load shall be distributed to the two posts as evenly as feasible. The applied load shall be 19,500 (10,000 x 1.95) pounds for a class 1 upright frame, 31,200 (16,000 x 1.95) pounds for a class 2 upright frame, and 39,000 (20,000 x 1.95) pounds for a class 3 upright frame. Before the load is applied, measurement reference points or lines shall be established on the upright posts of the frame, and the dimensions shall be noted for each reference point. Measurements shall be taken and noted 18 inches from the top and bottom of the upright posts, and 18 inches from the ends of diagonal or horizontal struts in the approximate center of the upright posts. The measurements shall include the overall depth of the upright frame, post and strut dimensions, and the width of any opening where the post formation is not a complete enclosure. The load shall be applied for 15 minutes. After 15 minutes, the upright frame shall be visually examined with the load applied, for metal rupture, weld or component failure. Evidence of any of the above defects shall be sufficient cause to reject the rack unit. In absence of any of the above defects, the load shall be removed and dimensions shall be rechecked for permanent deflection. Variation of 0.046 inch on the overall depth of the upright frame, 0.030 inch on dimensions where flanges terminate, including distance between terminal ends of flanges, and 0.020 inch on all other dimensions, shall be sufficient cause to reject the rack unit.

4.5.4 Shelf load test. The test unit shall be as specified in 4.5.1, and the unit shall not be anchored to the floor or to a wall. A test load of 1500 pounds (including containers) of sand, gravel, concrete, steel ingots, scrap steel, or steel test weights shall be placed, uniformly as feasible on the surface of the shelf. The distributed load may or may not be evenly distributed on the shelf. The individual weights shall be not more than 17 inches wide and the depth shall be not less than 1/3 or more than 1/2 of the depth of the shelf that is being tested. The weights may extend over the front and back of the shelf. One shelf of each bay of the test unit shall be subsequently tested. A load of 375 pounds shall be placed on each end of the shelf to be tested to "set" the shelf-to-post connection. The load, all except 25 pounds at each end of shelf, shall be removed and a dial indicator shall be set at each corner of the shelf, one at the approximate equidistant point from each shelf end, front and back, and an additional indicator placed under the approximate center of each shelf section, for determining the shelf deflection. The 1500 pound load shall be placed on the shelf for 15 minutes. At the end of 15 minutes, the deflection of the shelf shall

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be noted. Deflection of the shelf in excess of 0.400 inch, at any point tested, shall be sufficient cause to reject the rack unit. If the shelf deflection is less than 0.400 inch, an additional load of 975 (1500 x 1.65) shall be added to the existing load for 15 minutes. At the end of 15 minutes, the shelf shall be visually examined, with the load applied, for metal rupture, weld or component failure. Evidence of any of the defects stated above shall be sufficient cause to reject the unit. In absence of the above defects, the shelf shall be unloaded, all except 25 pounds at each end of shelf, and checked for permanent deflection. Permanent deflection of the shelf or the shelf supporting components in excess of 0.030 inch shall be sufficient cause to reject the rack unit.

4.5.5 Beam and shelf adjustability test. The test unit shall be as specified in 4.5.1, except heights 1 thru 4 that utilize tie beams shall consist of six (three pair) of tie beams. Two shelves of the test unit shall be loaded with 1500 pounds as specified in 4.5.4. The components of the unloaded shelf shall be removed from the unit and repositioned at a new location on the upright posts. One pair of tie beams shall be removed from the test unit, that utilizes tie beams, and repositioned at a new location on the upright posts. The components shall disassemble and assemble without the use of special tools, and the repositioning of the components should be accomplished using a soft hammer (lead, rawhide, hard rubber, etc.) weighing 2-1/2 pounds to 3 pounds and a screwdriver less than 9 inches long. The inability of any component to disassemble or assemble shall be sufficient cause to reject the rack unit.

4.5.6 Safety lock test. The beam-to-post, end support, and shelf safety locks shall be tested for effectiveness. The beam-to-post safety lock shall be tested by striking the beam, near the safety lock, twice on the top, side, and bottom with substantial blows using a 2-1/2 pound to 3 pound soft hammer specified in 4.5.5. Attempts shall be made to disengage the beam from the post, without disengaging the safety lock by striking the beam on the bottom with substantial blows using the above stated hammer. If the beam separates from the post, safety lock is ruptured, distorted, or fails to operate, the safety lock shall be considered as failing to meet the test. An end support (when applicable) shall be assembled to the upright frame, at a convenient height, and the safety lock shall be tested by striking the support twice on the bottom, sides, and top, near the ends of the support, with the above stated hammer. If the support separates from the post, safety lock is ruptured, distorted, or fails to operate, the safety lock shall be considered as failing to meet the test. One complete shelf (including sections, when applicable) shall be assembled, at a convenient height, to the assembled test unit. The shelf (and shelf sections) safety lock(s) shall be tested by striking the shelf or sections twice on the top, each side, and bottom. If the shelf or shelf sections separate from the end supports, or shelf beams (when applicable), safety lock is ruptured, distorted, or fails to operate, the safety lock shall be considered as failing to meet the test. The failure of any safety lock to meet the test shall be sufficient cause to reject the rack unit.

4.5.7 Stability test. The test unit shall be as specified in 4.5.1, except; only the units that are 120 inches (height 3) high or higher shall be subjected to the test; the shelves shall be uniformly positioned from the bottom to the top of the post, and class 1 units (10,000 pound upright frame) shall have 14 shelves (7 shelves per bay), class 2 units (16,000 pound upright frame) shall have 22 shelves (11 shelves per bay), and class 3 units (20,000 pound upright frame) shall have 28 shelves (14 shelves per bay). The shelves shall be positioned on corresponding elevations within the bays. All of the shelves, except the top shelves shall be loaded to capacity (1500 pounds) with the material specified in 4.5.4 or with steel coils or rods. The top shelves of class 1 and class 2 units shall be loaded with 1,000 pounds each, and the top shelves of the class 3 units shall be loaded with 500 pounds each. The load on the shelves shall be distributed as uniformly as possible. A 75-pound weight shall be attached to one end of a flexible line. The line shall pass over a 4-inch to 6-inch diameter pulley and attach to a post of the intermediate (common) upright frame, not farther

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than 3 inches from the shelf-to-post attachment. The pulley shall rotate freely on a shaft that is secured to a test fixture, an arm projecting from a wall, or to a rigid upright. The pulley shall be positioned so that the pull to the post is perpendicular to the longitudinal axis of the rack, and parallel to the supporting surface (floor) of the rack. A 75-pound weight, as described above, shall be attached to the same post at each level where the shelves are fully loaded. A 50-pound weight shall be attached to the post, at the partially loaded shelves of the class 1 and class 2 units, and a 25-pound weight shall be attached to the post, at the partially loaded shelves of the class 3 unit. The weights shall suspend freely from the pulley so that the maximum pull is exerted to the post. The attitude of the post, to which the weights are attached, shall be established prior to applying the lateral pull, by positioned dial indicators on two adjacent sides of the post, at the top of the post. The indicators shall be rigidly supported by a test stand, wall mount, or an upright similar to the test unit. The force exerted by all of the weights shall be applied to the post for 15 minutes. At the end of 15 minutes, the rack shall be visually examined for metal rupture, weld or component failure, and post deflection in excess of 0.025 inch per foot height. After the visual examination, the force shall be removed from the post, and after 5 minutes, the permanent deflection of the post shall be noted. Permanent deflection in excess of 0.008 inch per foot height or evidence of any of the above stated defects shall be sufficient cause to reject the unit. In absence of any of the above defects, each weight shall be attached, as stated above, to a post of the exterior frame. The post selected shall be on the same side of the test unit as the post selected above for testing. The dial indicators shall remain on the post of the intermediate upright. The lines shall be attached to the post so that the pull exerted is parallel with the longitudinal axis of the rack and parallel with the supporting surface of the rack. The force of all of the weights shall be applied to the post for 15 minutes. At the end of 15 minutes, the rack shall be visually examined for metal rupture, weld or component failure, and post deflection in excess of 0.075 inch per foot height. After the visual examination, the force shall be removed and after 5 minutes, the permanent deflection of the post shall be noted. Permanent deflection in excess of 0.020 inch per foot height, or evidence of any of the above stated defects shall be sufficient cause to reject the unit.

4.5.8 Back-to-back and wall-to-post connector test. The test unit shall consist of two style A (starter) units with 4 (two pair) of shelf or tie beams in each unit. The test unit may consist of the smallest height upright (see section 1) on order or contract, providing the connectors are compatible with the size of the posts of the upright. The units shall not be anchored to the floor. The beams shall be attached to the upright frames approximately 12 inches from the top and bottom of the frames. The units shall be positioned parallel and in alignment with each other and the upright frames on one end shall be joined together, top and bottom with the back-to-back connectors. The connectors shall be located 12 inches from the horizontal braces or 12 inches from the ends of diagonal braces. The outer post of one upright frame, on the same end of the unit that the back-to-back connectors are located, shall be attached to wall-to-post connectors. The attachment of the wall connectors shall be in horizontal alignment with the back-to-back connectors. The wall connectors shall be securely anchored to a steel, wood, or masonry wall or vertical barrier. A flexible line shall be attached to the outer post of the other outer post of the upright, run over a 4-inch to 6-inch diameter pulley specified in 4.5.7. A 250 pound weight shall be attached to the unattached end of the line so that the weight is at least 12 inches above the floor. A second line shall be attached to the weight to lift the weight 12 inches from its static position. After the weight has been lifted, the line shall be cut instantly to release the weight so that it can free fall to exert force against the connectors and upright frames. The direction of force shall be approximately parallel with the floor and perpendicular with the wall or barrier to which the wall connector is attached. The weight shall be raised and dropped three times. After the third drop, with the force applied to the upright frames and connectors, the connectors and upright frames shall be visually examined for component failure, metal rupture or distortion. Evidence of any of the above defects shall be sufficient cause to reject the rack units on contract or order.

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4.5.9 Finishing tests. A 20-gage steel panel shall be prepared in accordance with 3.7, and 2-inch wide test specimens cut from the panel shall be subjected to the following tests.

4.5.9.1 Flexibility test. The dry film shall show no evidence of cracking, or flaking under seven power magnification after the test specimen has been bent 180° over a 1/4-inch diameter mandrel.

4.5.9.2 Hardness test. The dry film on the test specimen and on the rack components shall withstand the firm stroke of a 2H pencil, held at a 45° angle and pushed across the film surface, without evidence of marring when viewed at an oblique angle in a strong light.

4.5.9.3 Adhesion test. The dry film shall not be removed from the test specimen and rack components when the film has been scored with a razor to the base metal, forming a grid of 1/8-inch squares, and a one inch wide piece of cellophane tape (Scotch Brand No. 600 or equal), is firmly applied to the grid surface and quickly pulled therefrom.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Racks shall be disassembled and packaged knocked down. Like components of racks shall be bundled together and each bundle secured with waterproof tape or annealed copper wire. Erection instructions shall be packaged in accordance with Method IC-3 of MIL-P-116. Hardware removed during disassembly shall be placed in a cotton cloth drawstring bag and the bag secured to one of the components.

5.1.2 Level C. Racks shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Racks, packaged as specified in 5.1, shall be packed in a crate conforming to type II, style A of PPP-C-650. Each level A bundle shall be prevented from abrading adjacent bundles by the use of cushioning material, the contents waterproofed, and the crate closed and strapped in accordance with the appendix of PPP-C-650.

5.2.2 Level B. Racks, packaged as specified in 5.1, shall be packed in a crate conforming to type II, style B of PPP-C-650. Each level A bundle shall be prevented from abrading adjacent bundles by the use of cushioning material, and the crate closed and strapped in accordance with the appendix of PPP-C-650.

5.2.3 Level C. Racks, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking. Marking shall be in accordance with 5.3.1 or 5.3.2 as specified (see 6.2).

5.3.1 Civil agencies. In addition to any special markings required by the contract or purchase order, all marking shall be in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special markings required by the contract or purchase order, all marking shall be in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The boltless, solid shelf racks covered by this specification are intended for storing bulky and packaged items in storerooms and warehouses.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, symbol, and date of this specification.
(b) Style, class, depth, and height rack units desired (see 1.2).
(c) Whether preproduction sample is not required (see 3.1).
(d) Whether color, other than green, color no. 24277 of Fed. Std. No. 595 is required (see 3.2.2.3).
(e) Whether color, other than the color of the rack, cadmium, chromium, or zinc is required for the safety lock (see 3.2.2.3).
(f) Whether back-to-back connectors should be furnished (see 3.6.5).
(g) Whether back-to-back connectors shall provide other than 4-inch row spacing (see 3.6.5).
(h) Whether wall-to-post connectors are required, and whether the spacing shall be other than 4 inches (see 3.6.6).
(i) Whether leveling shims shall be furnished (see 3.6.7).
(j) Whether identification marking is not required (see 3.9).
(k) Whether identification marking in accordance with MIL-STD-130 is required (see 3.9).
(l) Level of packaging and packing required (see 5.1 and 5.2).
(m) Whether the shipping containers shall be marked in accordance with 5.3.1 or 5.3.2.

6.3 Federal specification coverage. This Federal specification does not include all the styles, classes, depths, and heights of the commodity covered by the title of this specification, but is intended to cover the styles, classes, depths, and heights which are generally procured by the Federal Government.

6.4 Recommendations. It is recommended that all single bay (starter) units and single row arrangement of racks be anchored to the wall, ceiling, and floor of the housing structure. Also, it is recommended that all double row arrangement of racks be joined with back-to-back connectors and be anchored as stated above.

MILITARY INTEREST:

Custodians:

Army - GL
Air Force - 84

Review Activities:

Army - ME, SM

User Activities:

Navy - MC, SA, YD

GSA DC 73-8662

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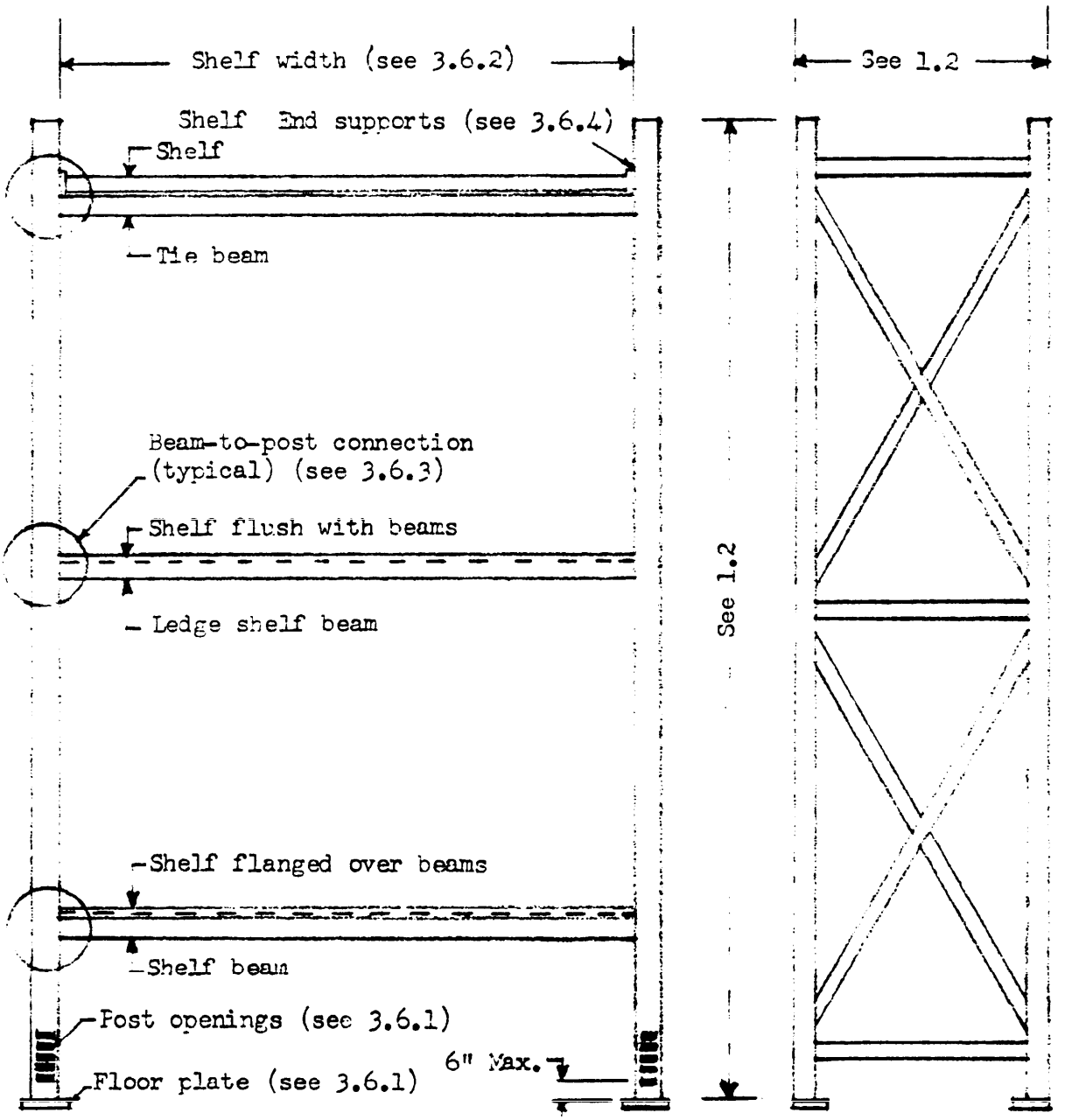


FIGURE 1. Typical Style A Storage Rack

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